AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM276
Software Development Processes and Methodologies

Final Examination
Fall Semester 2025/2026

Date: 20 /Jan./2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part I – Multiple Choice Questions — 5 Marks

*Answer **all** the following multiple-choice questions. Each question is worth 1 Mark.*

1. Which of the following is not a configuration management activity.
   a. N-Version development b. Version management c. System integration d. Problem tracking

2. Which testing strategy is where you identify groups of inputs that have common characteristics and should be processed in the same way.
   a. Guideline-based b. Unit c. Partition d. Automation

3. Which security dimension is meant to prevent the information from being damaged or corrupted.
   a. Confidentiality b. Integrity c. Availability d. Resilience

4. Which dependability property is best defined as "The ability of the system to deliver services when requested"?
   a. Availability b. Reliability c. Safety d. Security

5. Which security terminology is defined as "Possible loss or harm to a computing system"
   a. Asset b. Attack c. Exposure d. Threat

Part II – Short Answer Questions — 25 Marks

*Answer **all** the following short answer questions.*

1. Do you consider Inspection and Testing as Verification or Validation techniques? Are they complementary or opposing? (Make sure to explain)

   [6 Marks]

2. Lots of product companies are using an open-source approach to development
   a. List 2 advantages as a result of using an open-source approach.
   b. How do such companies make profit?
   c. Do such companies need to adopt techniques such as redundancy and diversity? Explain.

   [8 Marks]

3. In hazard assessment, a risk may be categorized as Intolerable, ALARP (As low as reasonably practical), or Acceptable. Define each of these 3 risk categories and provide an example for each based on the insulin pump case study you took in class.

   [6 Marks]

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4. Why is system integration a particularly critical part of the systems development process? Suggest two sociotechnical issues that may cause difficulties in the system integration process.

[5 Marks]

Part III – Problem Solving Questions — 20 Marks

*Answer **all** the following problem-solving questions. Each question is worth 10 Marks.*

1. Develop a class diagram for a system to be used at a resort with 4 main classes (Resort, Member, Membership, Activities). Make sure to represent the following:
   a. Include all 4 classes with proper associations and use Generalization to show that a VIPMember is-a Member (Note: 1 additional class is needed).
   b. Provide 5 associations as you find suitable and make sure that at least 1 of the associations is an aggregation association.
   c. Provide logical association names.
   d. Provide at least 8 multiplicity values.

2. a) Briefly discuss the reliability metrics:
   - POFOD (Probability of failure on demand)
   - ROCOF (Rate of occurrence of failures)
   - Availability

   b) Complete the following table by providing the appropriate reliability metric in each case and a suggested value. The first row was completed for you as an example.

System	Reliability Metric	Suggested Value
Patient monitoring system	Availability	System should be unavailable for less than 20 minutes per month
Word processor		
Vending machine controller		
Baking System controller		
Refrigeration unit control		
Management report generator		

End of questions

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